

FBSD PAC Assignment

Q1) Timing and power dissipation in FPGA-based sequential circuits.

Ans) Timing: Interconnected delay, not logic delay, dominates the critical path. Since a signal crosses several switch matrix stage before reaching its destination. This causes clock skew across the clock tree, setup/hold violations on congested routes, and glitches on combinational inputs that can be mismatched near a clock edge.

Power: dynamic power comes mainly from switching the programmable interconnect and clock network (the clock toggles every register every clock, used or not). Static power comes from leakage in the many idle configuration transistors.

Assignment.

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Q2) SoC, vs. conventional processor
system for a wearable health
monitor.

Ans) A conventional processor-based system
is simplest but every off-chip link
costs power, & continuous signal processing
competes with the processor's other tasks.
An SoC integrates everything on one
chip but is fixed at fabrication, so a
custom accelerator can't be added
or revised later.

SoPC is better fit: It combines a processor
with reconfigurable FPGA fabric, so the
custom hardware acceleration the device
needs, can be run continuously at
low power alongside the processor,
and can be updated post-fabrication
as algorithms are refined.

Q3) Hardware-software co-design for a Real-time Image Processing System.

Ans) Pixel level, regular, high parallel operation belong in hardware: convolution, filtering, edge detection, color-space conversion, thresholding, histograms, FFT/DCT. A pipelined hardware block can sustained one pixel per cycle which software cannot match.

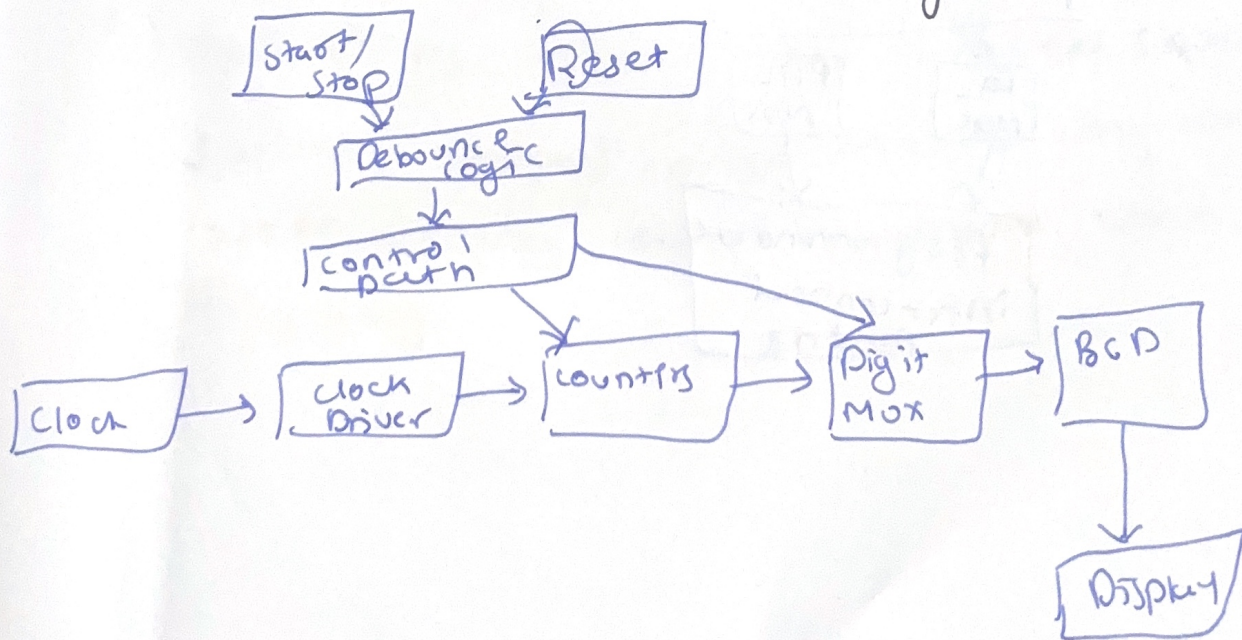
Co design lets the hardware pipeline handle the throughput-critical pixel work while the processor supervises & handles exceptions cutting latency versus a software-only design. Co-simulating both together verifies the partition before implementation.

Clock

Q4) Digital stopwatch as a top-down system design.

Ans) Top-down design starts from overall behaviour (start/stop/reset/display)

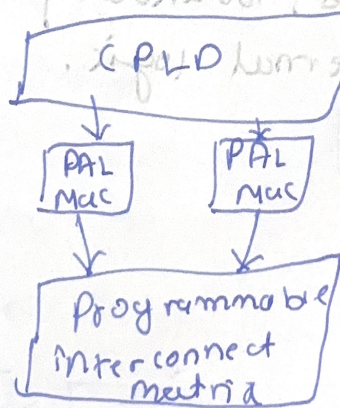
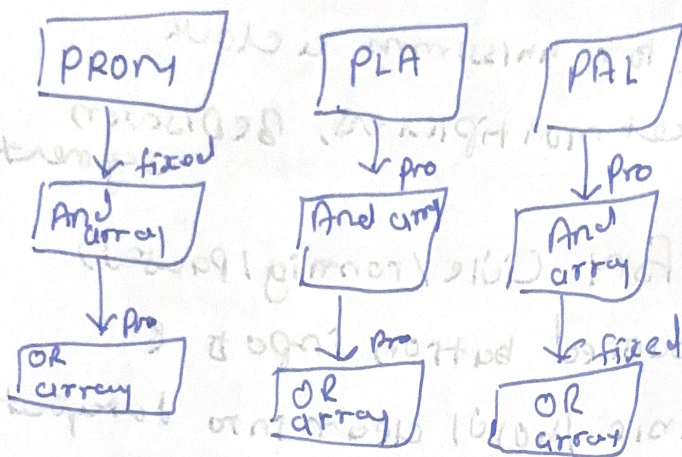
- Datapath: counter for milliseconds a clock driver, digit select multiplexers, BCD to seven segment.
- control path: an FSM (idle/running/paused) that reads deounced button inputs & drives count-enable/hold/clear into datapath
- user interface: buttons, boundary between user & internal logic.



Q5) PROM, PLA, PAL, and CPLD

architecture

Ans All four implement SOP logic via
And array & OR array.



Q6) programmable routing & interconnections:
= flexibility, area, power & delay.

- flexibility: more switches segments cause rout and raise utilization: too little routing makes design unroutable.
- Area: Internament typically takes up the majority of FPGA die area, so more flexibility costs significant area.
- Power: Each switch point add capacitance & a driver must charge/discharge, so longer; multihop routes draw more dynamic power, and idle switches leak.
- Delay: routing delay usually exceeds logic delay, since a signal crosses several switch stages; congestion forcing longer routes directly lowers maximum clock frequency.

(Q7) Altera Stratix vs Xilinx Virtex-11 routing for low power.

Ans) Both use segmented interconnect (multiple wire length plus a switch fabric) rather than one uniform resource, which helps limit power.

Stratix's multitrack interconnect adds direct link paths between adjacent logic array blocks (letting local connection skip the full switch matrix and use less capacitance). Virtex-11 routes

through a per-CLB switch matrix with direct, long & long lines, giving strong global flexibility, but at higher average power per connection when a local link could have been used instead.

Q5) SoPC advantages and hardware-software co-simulation for an embedded system

Ans) SoPC advantages:

single chip integration

on chip bus results low latency.

Accelerator logic.

work splits naturally.

new peripherals can be added later.

co-simulation.

Running the software (on an IIS) together with a simulated model of the hardware acceleration over a modeled bus, verifies data transfer interrupts & timing before synthesis. This catches interface bugs early and gives a performance estimate before committing to hardware.

9) programmable interconnection schemes & power dissipation

- crossbar / fully connected: maximum flexibility but switch count (power) grows roughly with the square of connection.
- Multiplexer-based: power rises with mux tree depth and fan-in.
- Switch-matrix with pass transistors (typical FPGA approach) power scales with the number of switch points a route passes through.
- segmented / hierarchical: matches wire length to connection distance cutting average power since most connections are local.
- Dedicated hard wired links lowest power, but no flexibility.

Q10) Hardware-software co-design strategy for an FPGA-based vending machine controller.

Ans) Hardware:

- coin/card sensor
- keypad
- Dispense-seq controller.
- Display driver.

Software

- price lookup & transaction logic
- Inventory tracking & restocking alert
- Display messages / prompt
- Transaction logging & remote communication
- Exception handling (out-of-stock etc)

